SMART INDIA HACKATHON 2025

TITLE PAGE

- Problem Statement ID – SIH25043
- Problem Statement Title- Embedded Intelligent Microscopy System for Identification and Counting of Microscopic Marine Organisms
- Theme- Smart Automation
- PS Category- Hardware
- Team ID- 51137
- Team Name- ARES

ARES

IDEA TITLE

SMART INDIA HACKATHON 2025

❖ IDEA / SOLUTION :

- LabInBox AI is a portable, intelligent device that attaches to any standard microscope.
- It uses **on-device fully offline** AI for real-time identification and counting of marine microorganisms, delivering instant results directly in the field.
- Provides the identification and counting for **127 different marine microorganisms**.
- Having **83% (mAP50-95 score)**average accuracy for identification of microorganisms.
- The model trained on **56,000 realistic, perfectly labeled training images**.

❖ Problem Resolution :

- Our solution directly replaces **the slow, expensive and error-prone manual counting process**.
- It provides an automated and portable system for **water surveys and research purposes** in marine science.

❖ Unique Value Propositions (UVP) :

- Combining cost-effective **Edge AI** for field deployment with our own **Synthetic Multi-Organism dataset** and **high-detail organism segmentation** to create a highly effective dataset for training a YOLOv8 object detection model.

ARES

TECHNICAL APPROACH

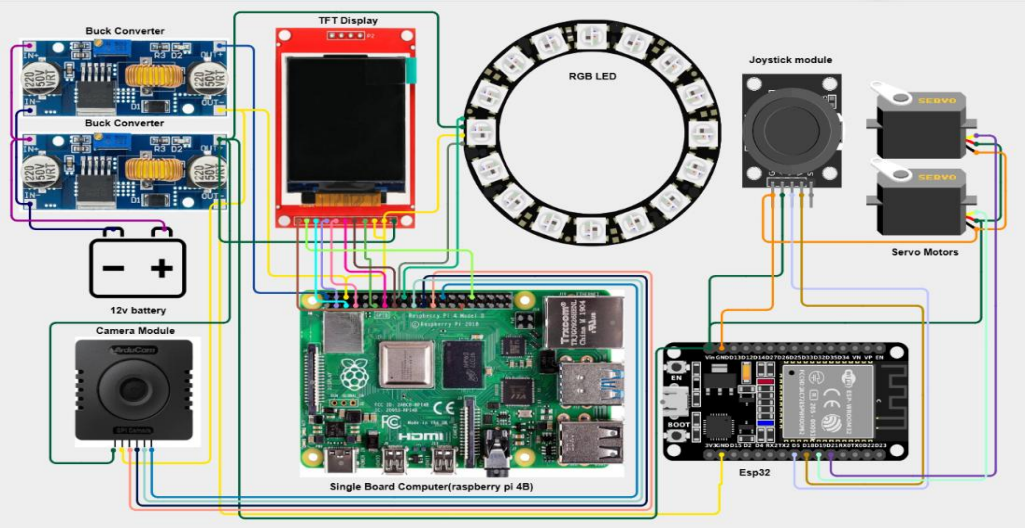
SMART INDIA HACKATHON 2025

- Our system uses **YOLOv8** computer vision model for training, optimized to run on a **Nvidia Jetson Nano** and **Raspberry pi**.
- RGB LEDs for various lighting conditions.
- **Joystick control** system for controlling live video feed.
- **On-device display** for real time detection and counting of microorganisms.
- We've created a **synthetic multilabel dataset** that includes thousands of realistic, perfectly labeled training images.

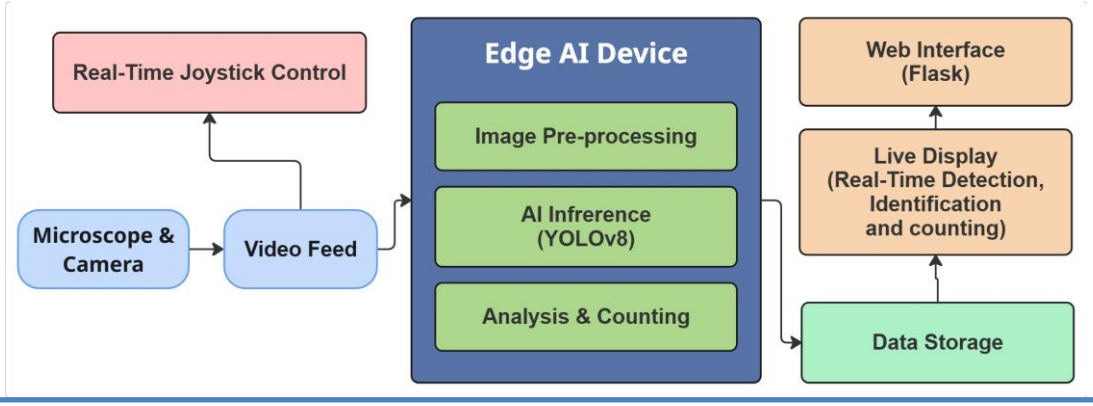
➢ **Algorithms**

- now we are used the **Synthetic data generation algorithm** by combining multiple organism images with artificial water noise.
- **GrabCut algorithm** for organisms image segmentation.

CIRCUIT DIAGRAM



PROCESS FLOW ARCHITECTURE



ARES

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2025

❖ Analysis of the feasibility of the idea

- The core technology is proven, all hardware components are readily available off-the-shelf.
- The market demand from **research, aquaculture and environmental monitoring sectors**.

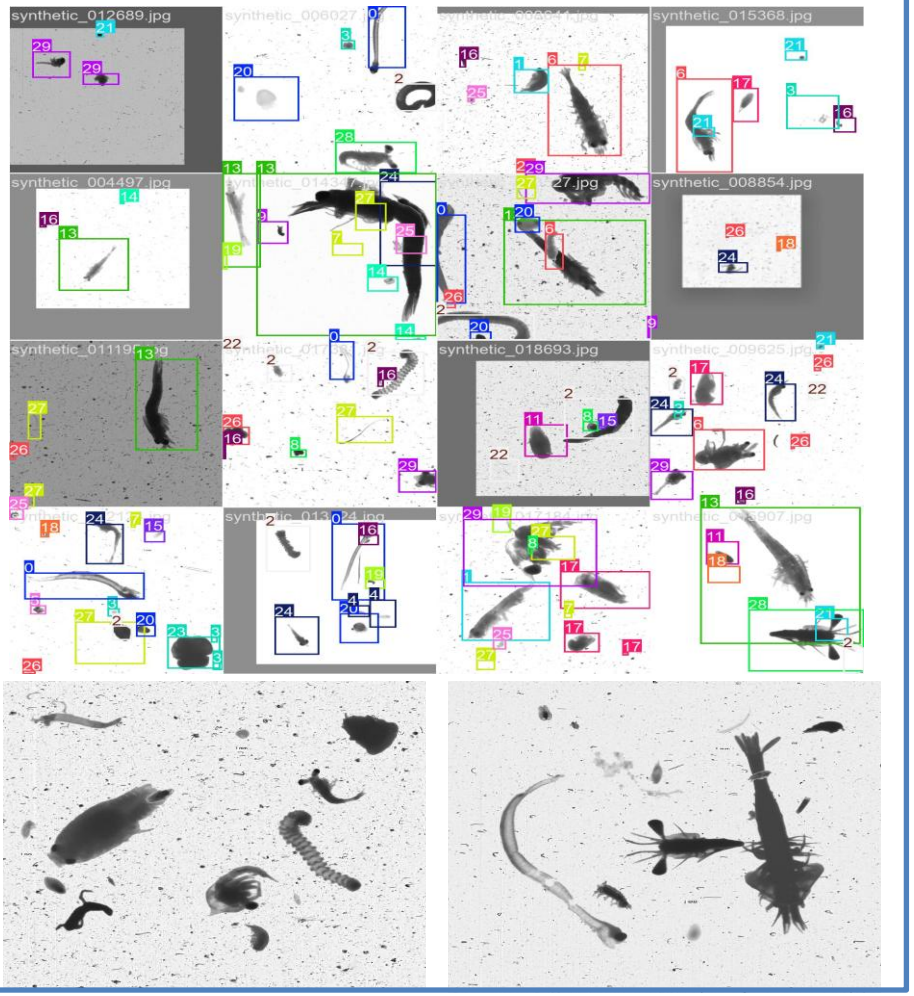
❖ Potential challenges and risks

- Identification for **overlapping organisms.**
- Handling highly **noisy water samples.**
- Ensuring model accuracy across diverse geographical regions.

❖ Strategies for overcoming these challenges

- Mitigate these risks through **Synthetic Multi-Organism dataset** generation, also solve overlapping organisms problem using **GrabCut algorithm.**
- Dataset includes realistic water noise & pre-processing filters to handle sample noise.

Synthetic Multi-Organism Dataset Sample Img.



ARES

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2025

❖ POTENTIAL IMPACT ON THE TARGET AUDIENCE

- This device will empower marine **biologists and environmental scientists** by transforming a long data analysis process into an instant, on-site task.
- Now the target audience for this product is **Marine biologists, environmental agencies, aquaculture managers, and academic institutions**.

❖ BENEFITS OF THE SOLUTION

- **Economic**: Lowers marine survey operational costs.
- **Environmental**: Enables early detection of pollution, invasive species, and harmful algal blooms.
- **Social**: Supports vital research on climate change, ocean health, and food security by providing better, faster data to the global scientific community.

ARES

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2025

Primary Datasets Used:

- SEANOE Plankton Imagery Dataset: https://www.seanoe.org/data/00829/94052/
- Our own Synthetic Multi Organism Datasets:
- https://www.kaggle.com/datasets/mohitpatel24/final-yolo-dataset-split
- https://kaggle.com/datasets/cd206da22c9575e61868c38bee40c2bdf2e686b4cc5650bf01062954b30f5b44

- YouTube video link: https://youtu.be/-DJN3FKYtXQ?si=cDiSmPnVjw0AMBV2

References

- Reference 1: https://arxiv.org/html/2505.18477v3
- Reference 2: https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2021.746297/full
- Reference 3: https://arxiv.org/html/2410.16695v2